

K-Field Linear Inertia Derivation

Cell3 Traversal Phase Energy, Self-Indexing, and the Origin of Linear Inertia

Principal Investigator: P. Dwayne Esterline
BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Krytronx, LLC

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Foundational statement

Linear kinetic energy is the self-index-normalized, strictly positive quadratic phase cost of continuous traversal through the three coupled Cell3 plane channels. Linear inertia is the resistance to changing that coherent traversal-rate state. The lateral rotating-mass response is the residual mixed term between the common Earth-CMB traversal and an imposed local traversal.

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Abstract

This paper develops a complete linear-inertia model from the physical Cell3 geometry, its reciprocal plane families, the address-space coupling matrix W , the physical phase-response tensor G_{phase} , and the self-indexing closure architecture. A physical velocity v maps exactly into a three-component Cell3 traversal rate $qdot = 2 B^{-1} v$. The associated phase scalar is $qdot^T W qdot = v^T G_{\text{phase}} v$. Both W and G_{phase} are positive definite, so every nonzero direction on the unit sphere has a strictly positive raw traversal cost. The supplied phase metric has eigenvalues 1.4293287535, 2.1125450264, and 8.8003364347; therefore no direction, great circle, or projective ray is a zero-cost path.

The raw cost is direction dependent. Self-indexing supplies the occupied projective direction and therefore supplies the exact normalization required to convert the anisotropic raw phase scalar into the isotropic kinetic-energy law $E/m = v^2/2$. Along Earth's CMB traversal vector, the speed is 369,812.5020 m/s, the ordinary kinetic energy per unit mass is 68.3806433 GJ/kg, the raw Cell3 phase scalar is 293.0619367 GJ/kg, and the direction-indexed normalization is $\beta_E = 0.2333317117$. The equality $\beta_E V_E^T G_{\text{phase}} V_E = |V_E|^2/2$ is exact to the numerical precision of the supplied matrices.

The quadratic form also explains the rotating-tube interaction. For total traversal $qdot_{\text{total}} = qdot_E + qdot_R$, the phase scalar separates into an Earth baseline self-term, a local-motion self-term, and a mixed term $2 qdot_R^T W qdot_E$. The single-cell lateral-force reduction uses this same mixed scalar with coefficient κ_F . The resulting framework places linear kinetic energy, momentum, force, rotational coupling, and lateral residual response in one Cell3 traversal-energy architecture.

Reader guide

The paper is organized as a self-contained instructional reference. Chapters 1 through 4 establish geometry, units, planes, and self-indexed addressing. Chapters 5 through 10 derive positive traversal energy and evaluate Earth-CMB motion. Chapters 11 through 15 derive linear inertia, momentum, force, and the lateral mixed term. Chapters 16 through 18 state experimentally useful consequences and the remaining microscopic localization boundary. The appendices provide complete matrices, all 27 node coordinates, plane data, eigenvectors, source lineage, and numerical verification.

1. Definitions, scope, and dimensional contract

The physical Cell3 is a three-dimensional oblique basis embedded in the ICRS frame. Its basis matrix B has columns a , b , and c . The address variable q is dimensionless. In the centered half-step registry used by the single-cell reduction, physical position is $x = B q / 2$. Consequently, B carries units of length, $qdot$ carries units of inverse time, and physical velocity carries units of length per time.

$$\begin{aligned} x &= (1/2) B q \\ qdot &= 2 B^{-1} v \\ v &= (1/2) B qdot \end{aligned}$$

The direct metric $M = B^T B$ has units of length squared. The reciprocal metric $R = M^{-1}$ has units of inverse length squared. The address coupling W also has units of length squared. The physical phase-response tensor G_{phase} is dimensionless because

$$G_{\text{phase}} = 4 B^T W B^{-1}.$$

The scalar $qdot^T W qdot$ therefore has units m^2/s^2 , identical to energy per unit mass. The same scalar expressed in physical coordinates is $v^T G_{\text{phase}} v$. No dimensional conversion is hidden in this identity; the factor of four follows directly from the half-step coordinate convention.

Source-derived quantities and derived consequences

The matrices B , M , R , W , and G_{phase} ; the 27 centered nodes; the plane-family normals and spacings; the Earth-CMB vector; the rotating-tube parameters; the address rates; and the lateral-force reduction are supplied data. Eigenvalues, directional maps, kinetic normalization, phase decomposition, and the linear-inertia rules in this report are deterministic algebraic consequences calculated from those supplied quantities.

Physical and address-space forms

Object	Definition	Units	Role
B	Physical Cell3 basis	m or nm	Maps address displacement into ICRS displacement
q	Half-step address coordinate	dimensionless	Indexes the three coupled plane channels
$M = B^T B$	Direct metric	m^2	Physical squared length in address coordinates
$R = M^{-1}$	Reciprocal metric	m^{-2}	Plane normals, dual spacing, and reciprocal closure
W	Address coupling	m^2	Quadratic and mixed phase interaction in qdot-space
G_phase	$4 B^T W B^{-1}$	dimensionless	Physical-direction phase-response metric
C(u)	$u^T G_{\text{phase}} u$	dimensionless	Raw cost index for unit direction u
beta(u)	$1/[2 C(u)]$	dimensionless	Self-indexed kinetic normalization

2. Physical Cell3 geometry

The authoritative physical basis is expressed in nanometres. Its columns are the primitive translations associated with channels a, b, and c. The edge lengths are 0.3341796791 nm, 0.5864108904 nm, and 0.7923197650 nm. Their included angles are 83.933483 degrees between a and b, 80.448130 degrees between a and c, and 50.363232 degrees between b and c. The physical cell volume is 0.1179148919 nm³.

B (nm)	col 1	col 2	col 3
row 1	0.32750124207	0.0353307758441	-0.0101293176267
row 2	-0.0548970078863	0.207319850441	-0.389787702364
row 3	-0.0374875028963	-0.547401085392	-0.689734408363

Physical Cell3 half-step registry: all 27 nodes

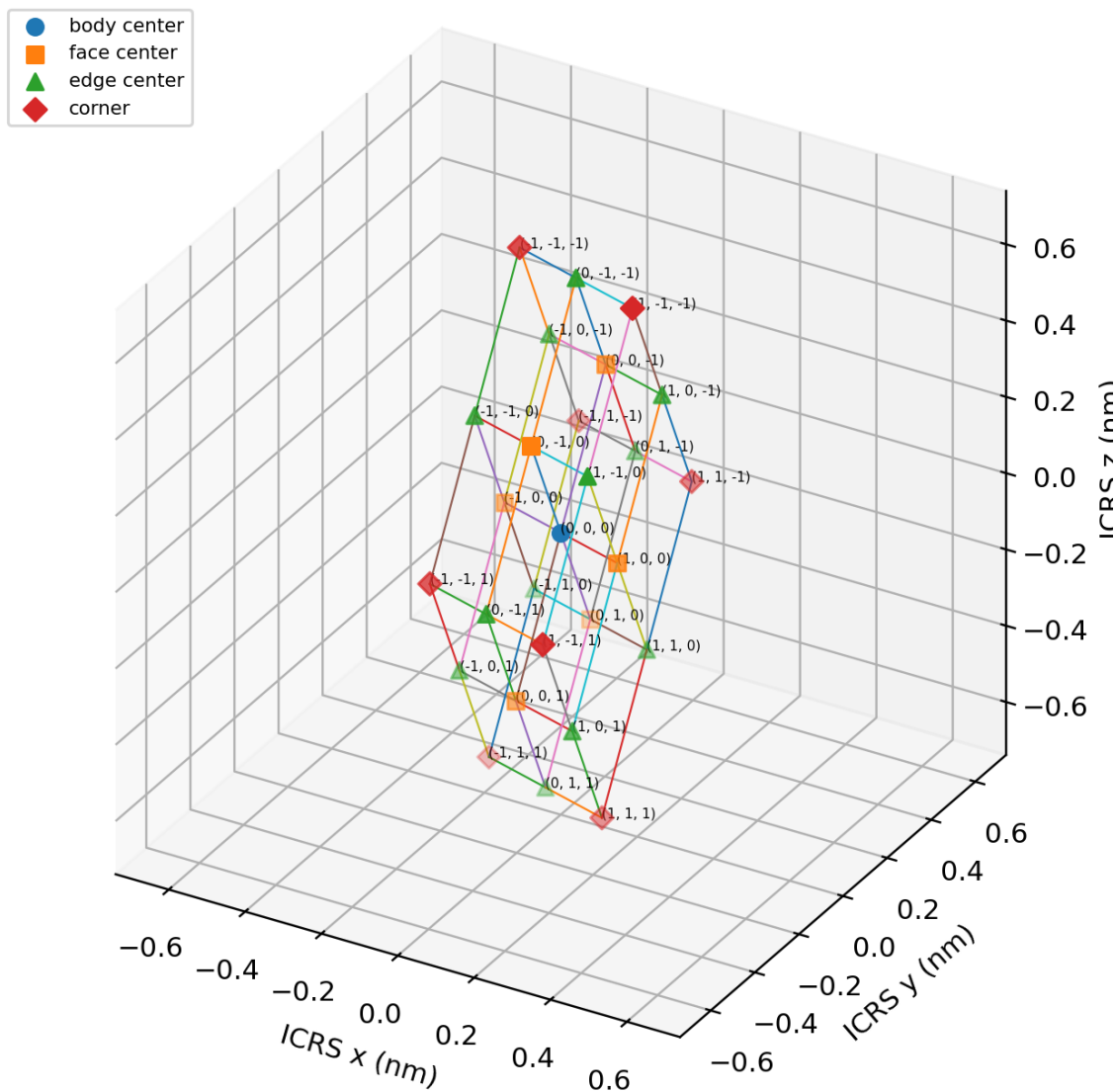


Figure 1. The complete 3 x 3 x 3 centered half-step registry. Every one of the 27 supplied nodes is plotted at its physical ICRS coordinate and labeled by its integer address L in $\{-1, 0, 1\}^3$. Adjacent nodes are connected only when their addresses differ by one unit in exactly one coordinate. All three axes use one common nanometre scale and orthographic projection.

Cell3 orthographic dimensional views

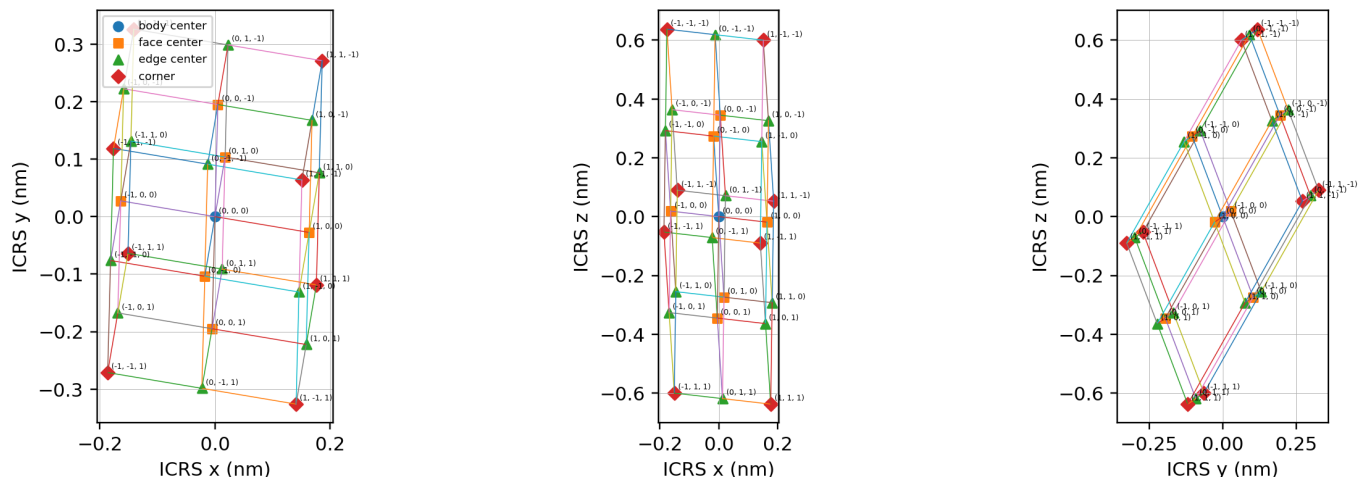


Figure 2. Orthographic x-y, x-z, and y-z projections of the same 27-node registry. These views remove perspective distortion and make the oblique physical geometry directly measurable.

Node rule and local registry

$$p_L = (1/2) B L, \quad L \text{ in } \{-1, 0, 1\}^3.$$

The center $L = (0,0,0)$ is the body center. Six addresses with one nonzero component are face centers; twelve addresses with two nonzero components are edge centers; and eight addresses with three nonzero components are corners. Antipodal addresses L and $-L$ occupy opposite physical positions. This centered construction is the local finite witness of the translation-invariant lattice $\Lambda = B Z^3$.

3. The three reciprocal plane channels

Each address coordinate measures traversal through one family of planes. The a channel is normal to the bc face family, the b channel is normal to the ac family, and the c channel is normal to the ab family. In the centered half-step registry, $q_i = -1, 0$, and $+1$ are three adjacent planes. Their separation is one-half of the primitive boundary-to-boundary spacing.

Channel	Face family	Unit normal in ICRS	Primitive spacing (nm)	Half-step spacing (nm)	Face area (nm ²)
a	bc	-0.995965190, 0.083602163, -0.032619291	0.329546529709	0.164773264855	0.357809539129
b	ac	0.089052594, 0.866576769, -0.491033948	0.451597379724	0.225798689862	0.261106235800
c	ab	0.194092201, 0.913177169, 0.358379235	0.605097741515	0.302548870757	0.194869165470

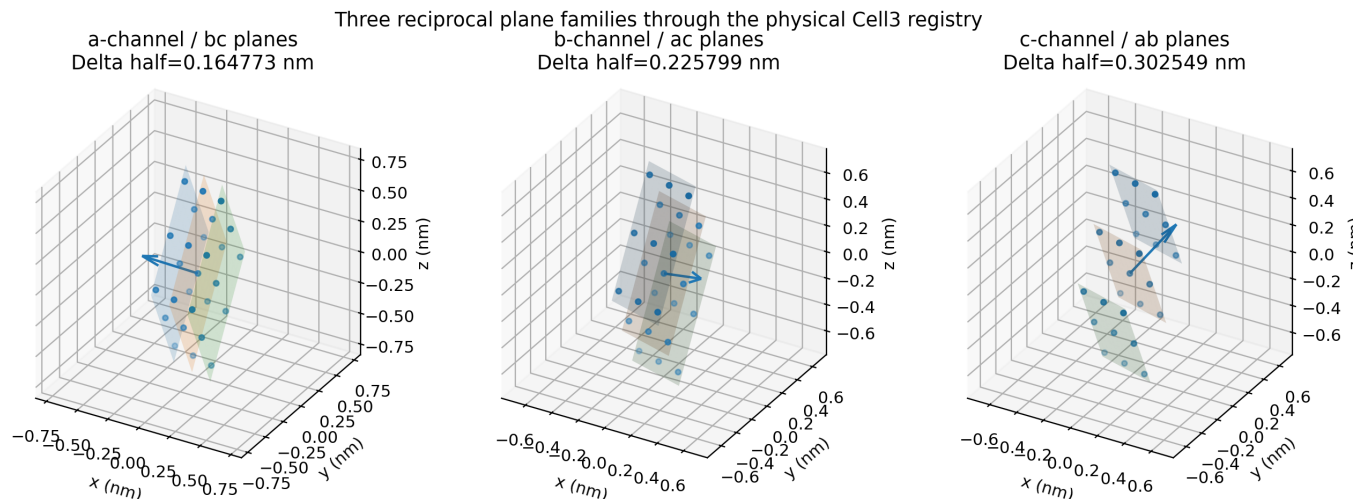


Figure 3. The a, b, and c reciprocal plane families rendered from the physical basis. Each panel contains the $q_i = -1, 0$, and $+1$ half-step planes, the complete node registry, and the supplied ICRS unit normal. Geometry and spacing are dimensionally correct in nanometres.

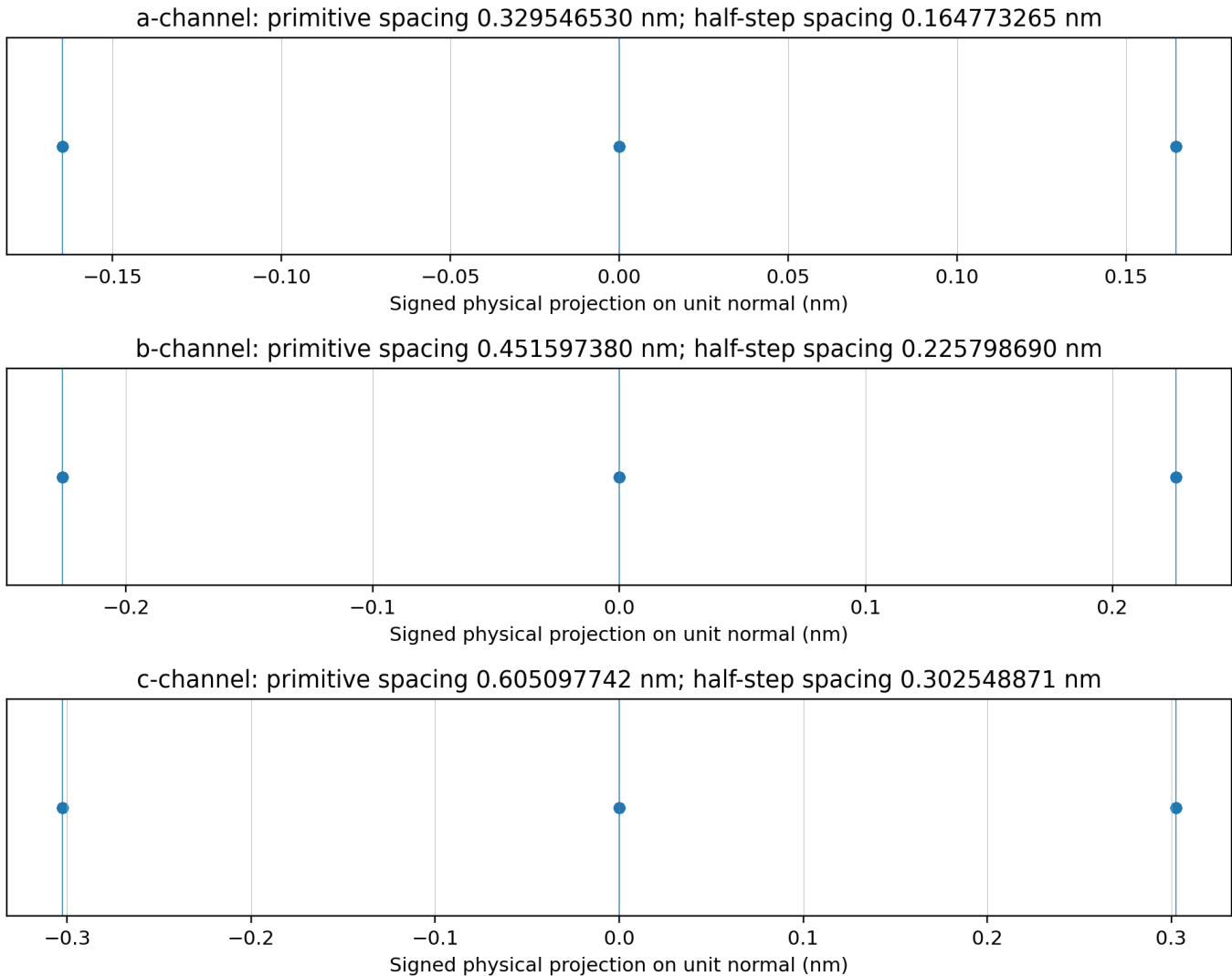


Figure 4. Signed projection of all 27 nodes onto each reciprocal plane normal. Repeated projected values identify coplanar nodes. The visible level separation equals the supplied adjacent half-step spacing.

Why every motion has a channel representation

Because B is nonsingular, B^{-1} exists. For every nonzero physical velocity v , $\dot{q} = 2 B^{-1} v$ is nonzero. A velocity may be tangent to one plane family and therefore have zero rate in one coordinate. It may also be arranged to eliminate two selected coordinate rates only when it lies along the remaining primitive direction. It cannot make all three rates vanish unless v itself is zero. This is the first structural reason that no physical direction escapes Cell3 traversal.

4. Self-indexing space and directional identity

Cell3 does more than supply three coordinates. The self-indexing architecture assigns each displacement a projective direction, a radial level, and an antipodal state. For an integer address k , write $k = s u$, where u is primitive and s is a positive integer. The directed angular state identifies u ; its projective version identifies the pair $\{u, -u\}$; the radial factor s identifies the level on that ray; and the antipodal polarity selects one direction.

$$k = s u$$

$$R(k) = \sqrt{k^T G_{\text{native}} k} = s \sqrt{u^T G_{\text{native}} u}.$$

Translation invariance means that the same local displacement dictionary is available from every lattice center. A velocity direction therefore carries a stable projective channel ratio independent of which particular Cell3 origin is selected. This property is essential to linear inertia: the energy law must move with the translating mass rather than depend on one privileged local cell index.

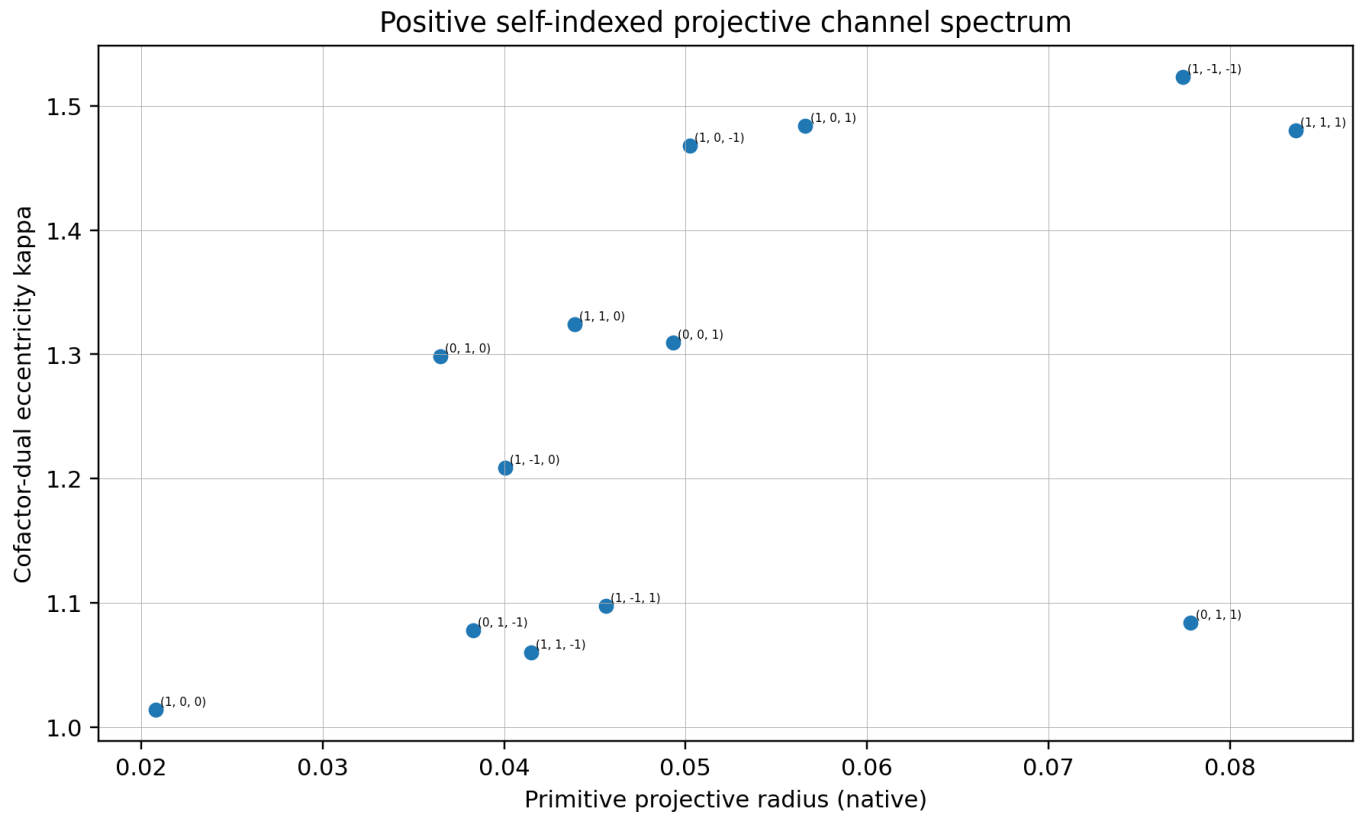


Figure 16. The 13 supplied Lucas projective channel representatives. Primitive radius and cofactor-dual eccentricity are both positive for every represented direction. Labels are the primitive addresses.

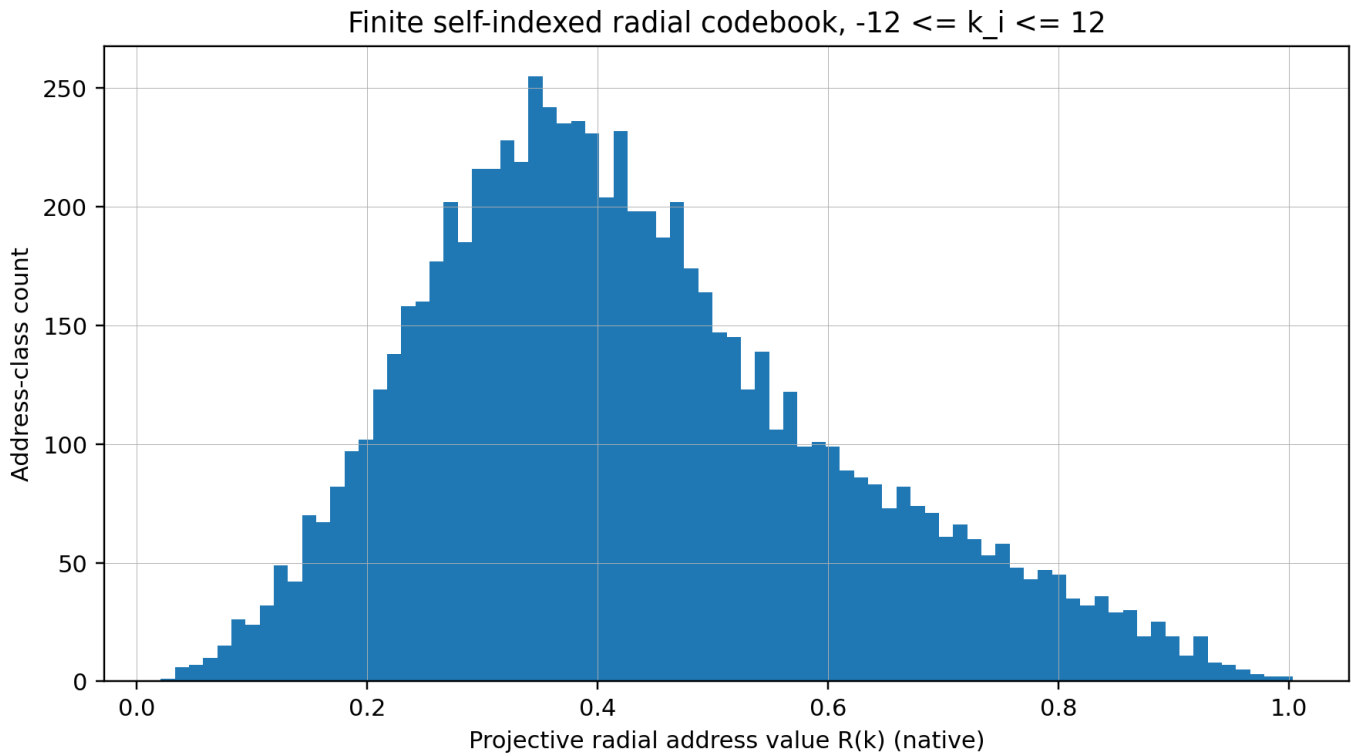


Figure 17. Finite projective radial codebook generated from the supplied native Gram metric over $-12 \leq k_i \leq 12$. The plot contains one representative from each antipodal pair and illustrates the discrete radial address spectrum used by self-indexing closure.

Closure basins and stable directional indexing

A measured witness may move continuously while the decoded address remains fixed inside its closure basin. The decoder requires simultaneous agreement of integer radial closure, primitive projective ray, antipodal state, direct syndromes, reciprocal syndromes, and global orientation. For inertia, this means that small non-coherent perturbations of the witness do not continuously rewrite the projective direction used by the normalization. The direction index is piecewise stable, while a genuine coherent transition carries the state into another complete closure basin.

5. Velocity as a Cell3 traversal-rate state

The half-step velocity equation supplies the exact bridge between ordinary motion and interior Cell3 phase interaction. Physical velocity is not represented by a single crossing count. It is represented by the simultaneous signed rates through all three plane families.

$$\dot{q}_a, \dot{q}_b, \dot{q}_c = \text{components of } 2^{-1} \mathbf{v}.$$

The signs specify which direction the material point crosses each family. The magnitudes specify the rate in half-step address units per second. Because the basis is oblique, the three coordinate rates are not orthogonal physical velocity components. Their physical meaning emerges only after contraction with the direct metric $\mathbf{M}$ or the coupling matrix $\mathbf{W}$.

Physical speed from address rate

$$|\mathbf{v}|^2 = \mathbf{v}^T \mathbf{v} = (1/4) \dot{\mathbf{q}}^T \mathbf{B}^T \mathbf{B} \dot{\mathbf{q}} = (1/4) \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}}.$$

This equation is the metric definition of speed in Cell3 coordinates. It also establishes the ordinary kinetic energy before any phase interpretation is introduced:

$$E_{\text{kin}} / m = (1/2) |\mathbf{v}|^2 = (1/8) \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}}.$$

The direct metric therefore defines the geometric rate cost, while $\mathbf{W}$ defines the phase-interaction rate cost. Their ratio along the occupied projective direction is the self-index used below.

6. Positive phase energy over the complete unit sphere

Define the raw Cell3 traversal phase scalar

$$\Phi(\dot{q}) = \dot{q}^T W \dot{q} = v^T G_{\text{phase}} v.$$

For a motion $v = s u$ with unit direction u and speed s ,

$$\Phi(v) = s^2 C(u), \quad C(u) = u^T G_{\text{phase}} u.$$

A symmetric matrix defines a strictly positive quadratic form exactly when all of its eigenvalues are positive. The supplied W and G_{phase} satisfy this condition.

Matrix	Eigenvalue 1	Eigenvalue 2	Eigenvalue 3	Condition ratio
$W \text{ (nm}^2\text{)}$	0.050269586792	0.094205549207	1.219025861836	24.249768889
G_{phase}	1.429328753487	2.112545026447	8.800336434665	6.156971525

Accordingly, for every unit direction u ,

$$1.429328753487 \leq C(u) \leq 8.800336434665.$$

There is no zero-cost direction, no zero-cost great circle, and no direction that can be removed from the phase-energy architecture. The all-direction positivity is an exact spectral property of G_{phase} . The maximum raw phase cost per squared speed is 6.15698 times the minimum.

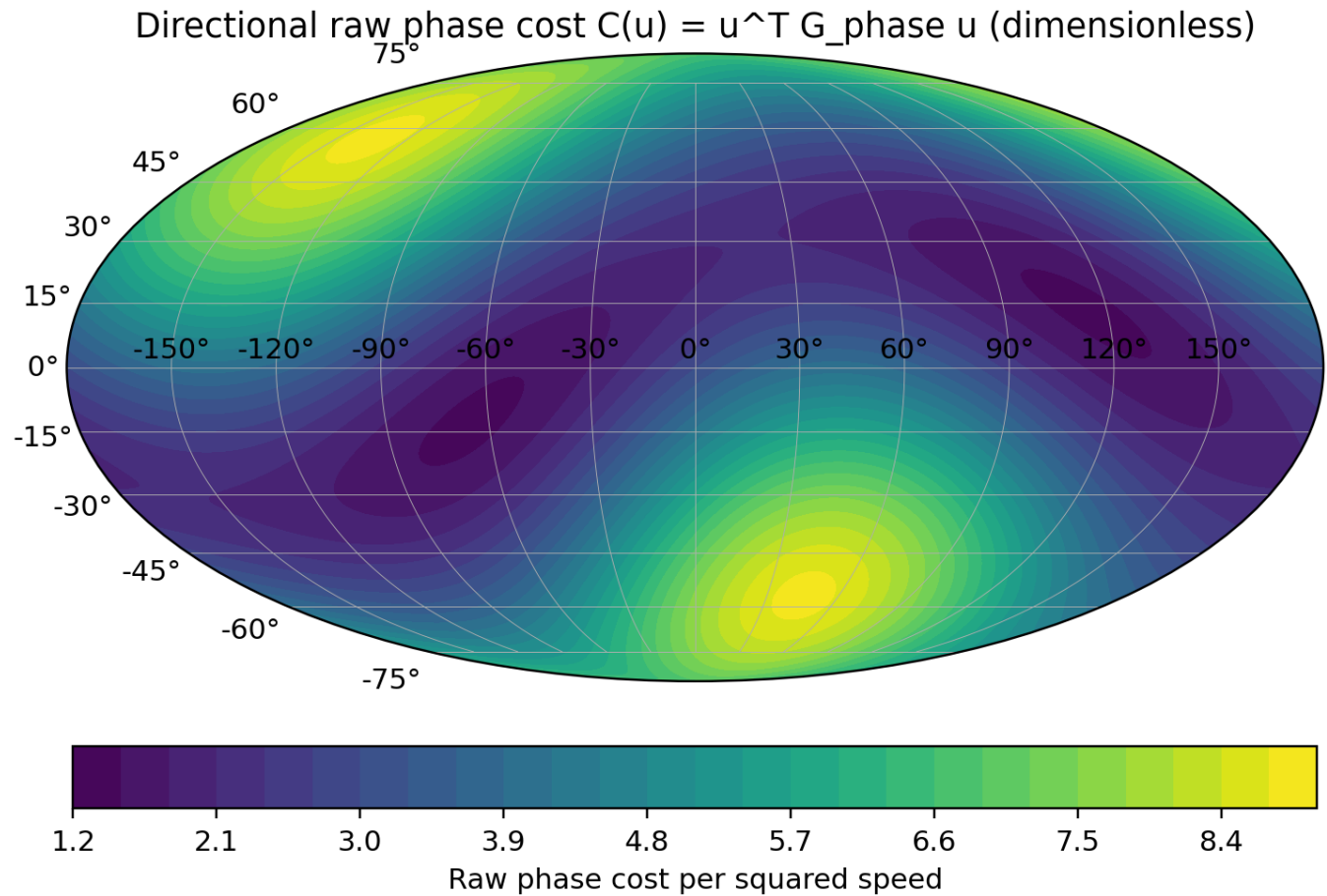


Figure 5. The complete unit-sphere directional cost $C(u)$. Every plotted value is positive. The map is antipodally even because $C(-u) = C(u)$.

Unit phase-energy ellipsoid: $x^T G_{\text{phase}} x = 1$

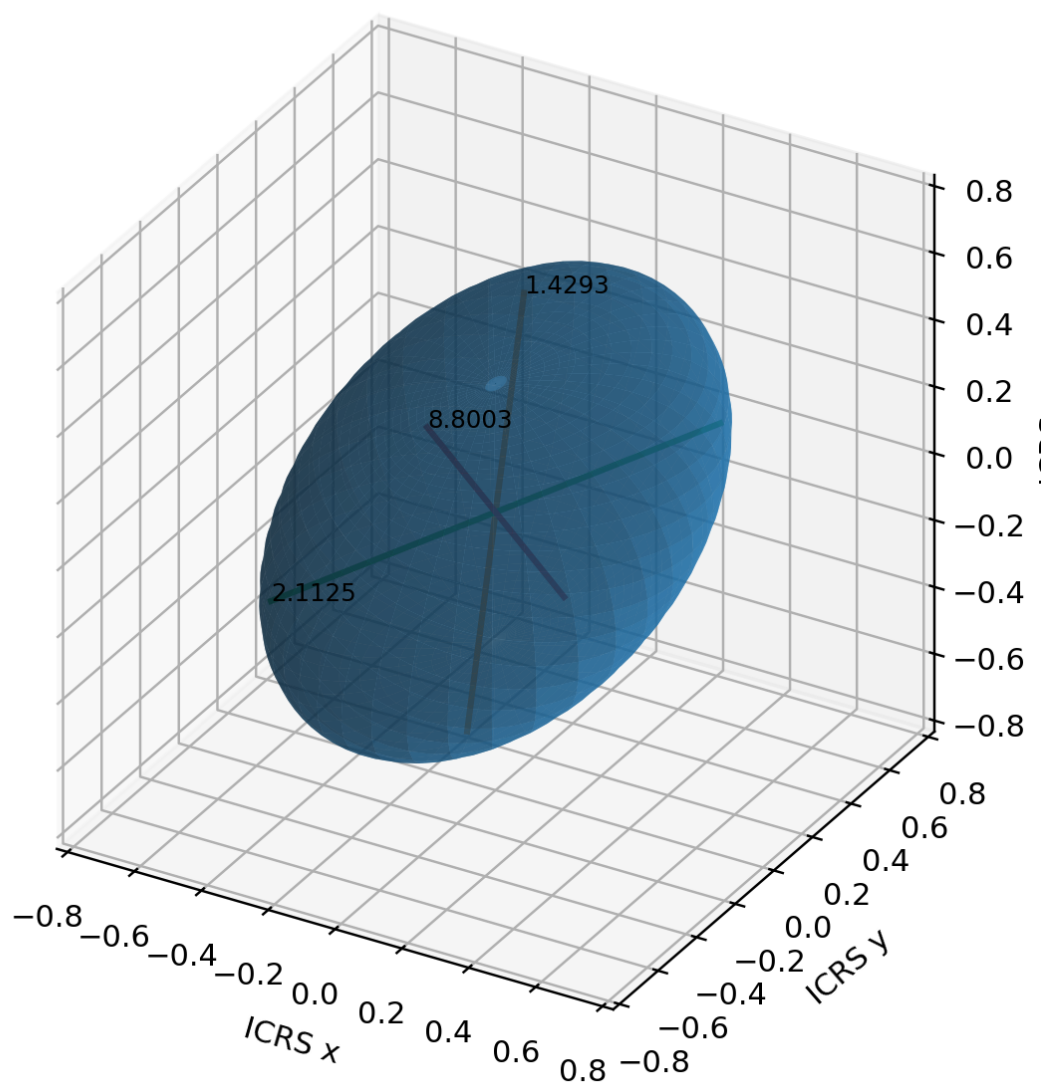


Figure 6. The dimensionless unit phase-energy ellipsoid $x^T G_{\text{phase}} x = 1$. Its principal semi-axes are $1/\sqrt{\lambda_i}$. The labels give the three eigenvalues of G_{phase} .

Antipodal parity

Since the phase scalar is quadratic, reversing velocity reverses every $\dot{q}$ component but leaves energy unchanged: $\Phi(-\dot{q}) = \Phi(\dot{q})$. The energy sector is therefore antipodally even. Its gradient is odd, so momentum and force reverse when the traversal direction reverses. This even-odd separation connects the quadratic energy form to the supplied cosine/sine channel decomposition of the spectral fabric.

7. Directional eigenstructure and phase anisotropy

The eigenvectors of G_{phase} are the physical directions in which the raw cost is purely diagonal. They form an orthonormal ICRS frame. A unit velocity along eigenvector e_i has raw phase cost $\lambda_i s^2$. Any other direction is a weighted mixture of these three principal responses.

Mode	Eigenvalue	ICRS unit vector	RA (deg)	Dec (deg)	Inverse-cost semiaxis
e1	1.429328753487	-0.408953067284, 0.884525009431, 0.224439070683	114.813083651	12.969895704	0.836438346450

Mode	Eigenvalue	ICRS unit vector	RA (deg)	Dec (deg)	Inverse-cost semiaxis
e2	2.112545026447	-0.828553895965, -0.256832305478, -0.497529504999	197.222212119	-29.836687522	0.688013583931
e3	8.800336434665	-0.382434086139, -0.389426083555, 0.837909001746	225.519007199	56.919969623	0.337093487547

Raw anisotropy is not itself anisotropic observed inertia. It is the direction-dependent internal phase gain that self-indexing must account for. The projective direction is available from the channel ratio $\dot{q}_a:\dot{q}_b:\dot{q}_c$; therefore the required normalization is an internal state function, not an external lookup based on celestial coordinates.

8. Earth-CMB traversal: complete numerical evaluation

The supplied Earth-CMB vector is

```
V_E = (-359003.199065813, 76679.886034002, -44701.060100688) m/s
|V_E| = 369812.502000406 m/s
RA = 167.943299880 deg, Dec = -6.942599922 deg.
```

The corresponding Cell3 address rates are

```
qdot_E = (-2.217734842862322e15,
          2.499065863112239e14,
          5.181751775713256e13) s^-1.
```

Direct substitution verifies $v = B \dot{q}_E / 2$ to a maximum component residual of approximately $5.8e-11$ m/s, which is at floating-point scale relative to the 369.8 km/s vector.

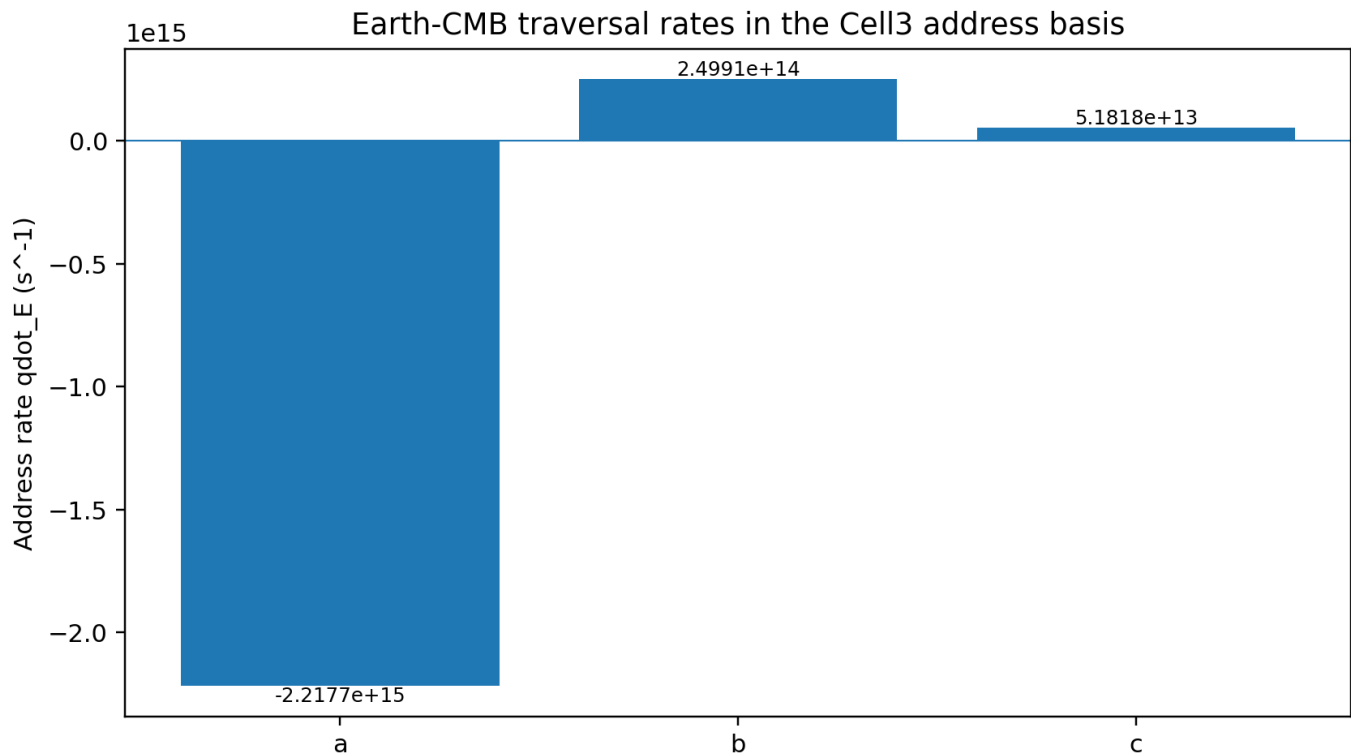


Figure 8. Signed Earth-CMB half-step address rates. Channel a dominates in magnitude and is traversed in the negative address direction; channels b and c are traversed in the positive direction.

Ordinary kinetic energy per unit mass

$$\begin{aligned}\epsilon_{\text{kin}} &= (1/2) |\mathbf{V}_E|^2 \\ &= 6.838064331790031 \times 10^{10} \text{ J/kg} \\ &= 68.38064331790031 \text{ GJ/kg}.\end{aligned}$$

Raw phase traversal scalar

$$\begin{aligned}\Phi_E &= \mathbf{V}_E^T \mathbf{G}_{\text{phase}} \mathbf{V}_E \\ &= \dot{\mathbf{q}}_E^T \mathbf{W} \dot{\mathbf{q}}_E \\ &= 2.930619366939657 \times 10^{11} \text{ J/kg} \\ &= 293.0619366939657 \text{ GJ/kg}.\end{aligned}$$

The equality of physical and address-space forms is a direct numerical verification of $\mathbf{G}_{\text{phase}} = 4 \mathbf{B}^T \mathbf{W} \mathbf{B}^{-1}$.

Earth-CMB traversal, phase-response direction, and phase eigenaxes

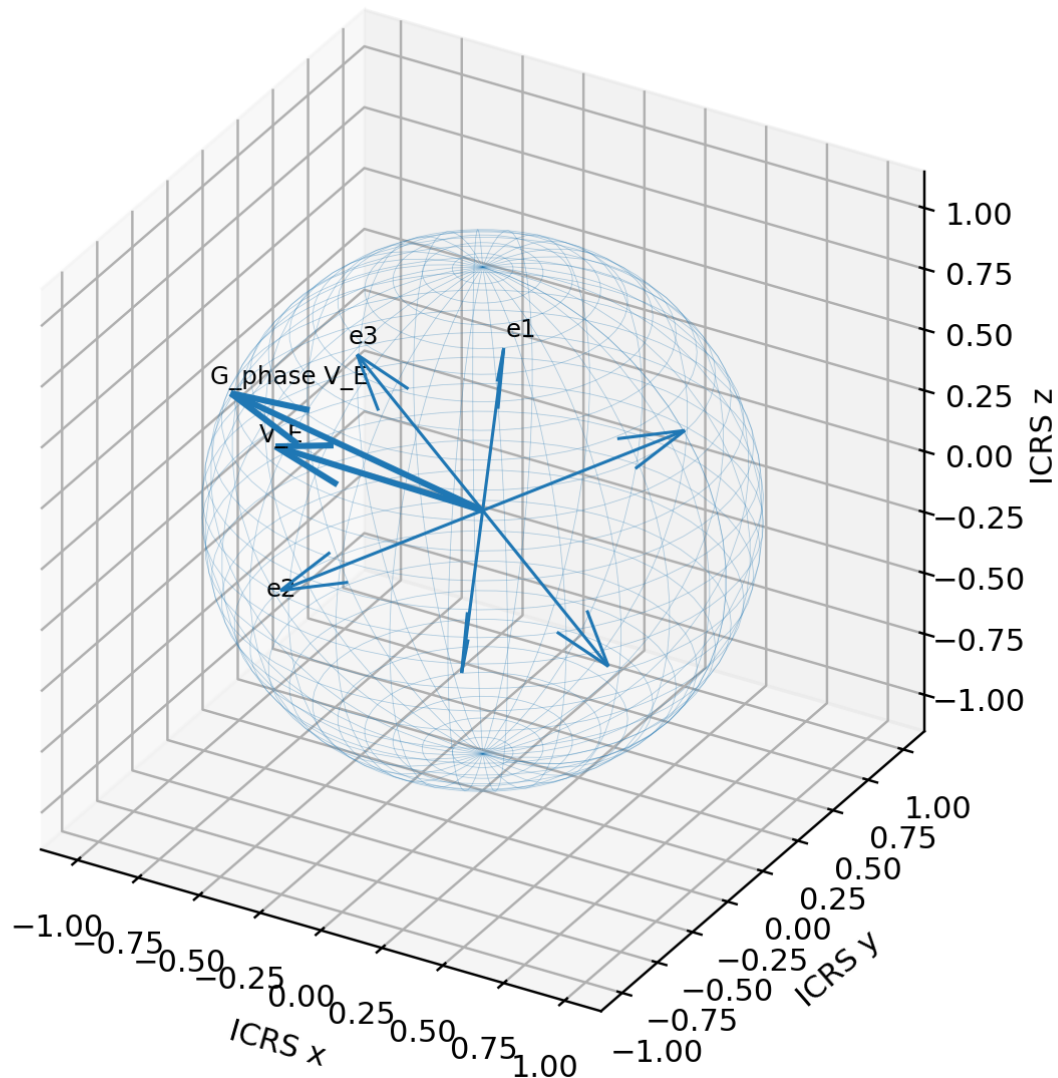


Figure 7. Unit directions of Earth-CMB traversal, the transformed phase vector $\mathbf{G}_{\text{phase}} \mathbf{V}_E$, and the three phase eigenaxes. The phase-response direction is not generally parallel to the physical velocity because $\mathbf{G}_{\text{phase}}$ is anisotropic.

9. Self-indexed normalization and recovery of kinetic energy

The raw directional phase index is

$$C(u) = u^T G_{\text{phase}} u.$$

In address coordinates, compare the phase metric W with the direct metric M along the occupied rate direction:

$$\chi([\dot{q}]) = (\dot{q}^T W \dot{q}) / (\dot{q}^T M \dot{q}).$$

Using $\dot{q}^T M \dot{q} = 4 v^2$, one obtains $C(u) = 4 \chi([\dot{q}])$. Therefore

$$\begin{aligned} E_{\text{kin}} / m &= (1/8) \dot{q}^T M \dot{q} \\ &= [1 / (8 \chi([\dot{q}]))] \dot{q}^T W \dot{q} \\ &= [1 / (2 C(u))] v^T G_{\text{phase}} v. \end{aligned}$$

Define $\beta(u) = 1/[2 C(u)]$. Then the normalized phase energy is exactly

$$\epsilon_K(v) = \beta(u) v^T G_{\text{phase}} v = (1/2) v^2.$$

For Earth-CMB traversal,

$$\begin{aligned} C_E &= 2.142872035668970 \\ \chi_E &= 0.535718008917242 \\ \beta_E &= 0.233331711682871 \\ \beta_E \Phi_E &= 68.380643317900 \text{ GJ/kg}. \end{aligned}$$

The raw phase tensor is anisotropic, but the self-indexed ratio between W and M returns an isotropic physical kinetic-energy scalar. The direction label and its normalization are two parts of one self-indexing state.

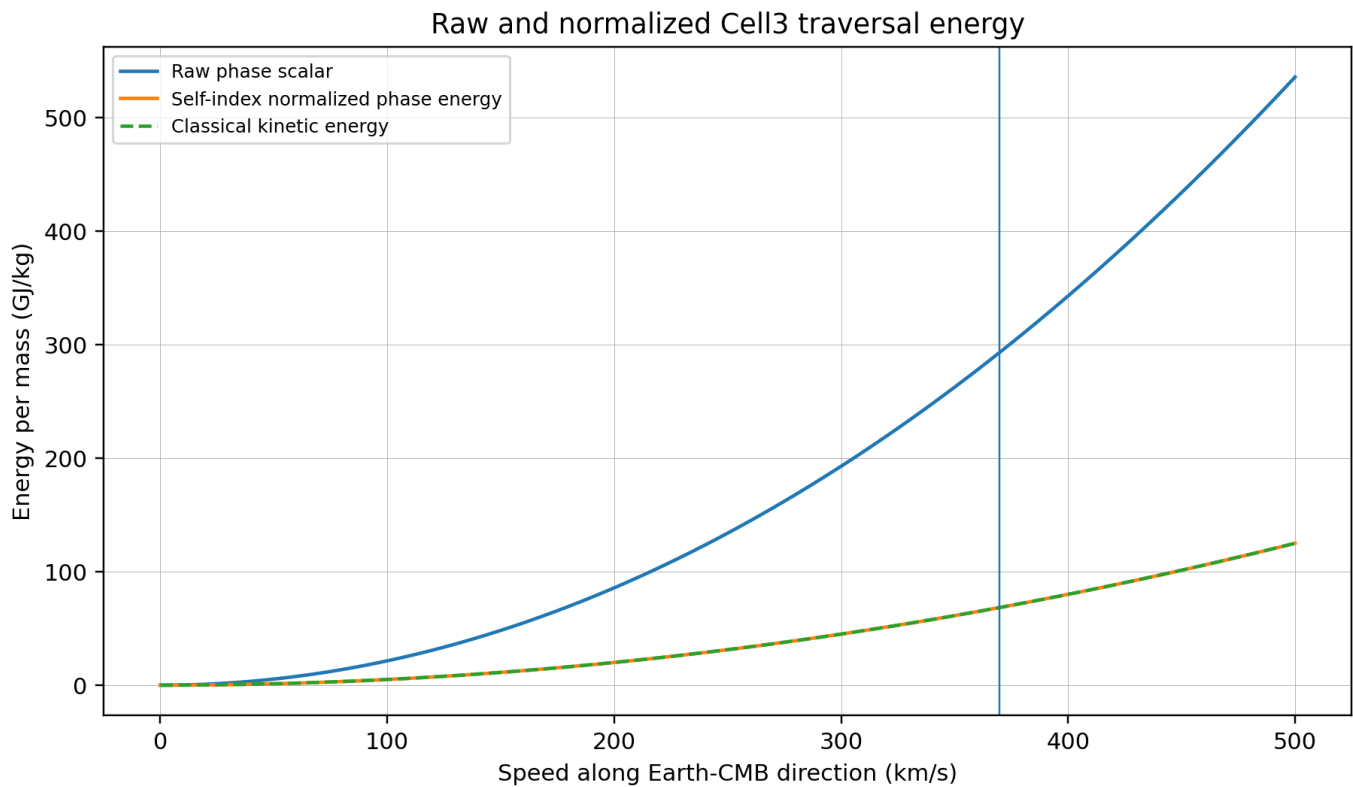


Figure 10. Energy per unit mass along the Earth-CMB projective direction. The raw phase scalar is larger than ordinary kinetic energy by the fixed factor $C_E/(1/2)$. Multiplication by β_E makes the normalized phase curve coincide exactly with $v^2/2$.

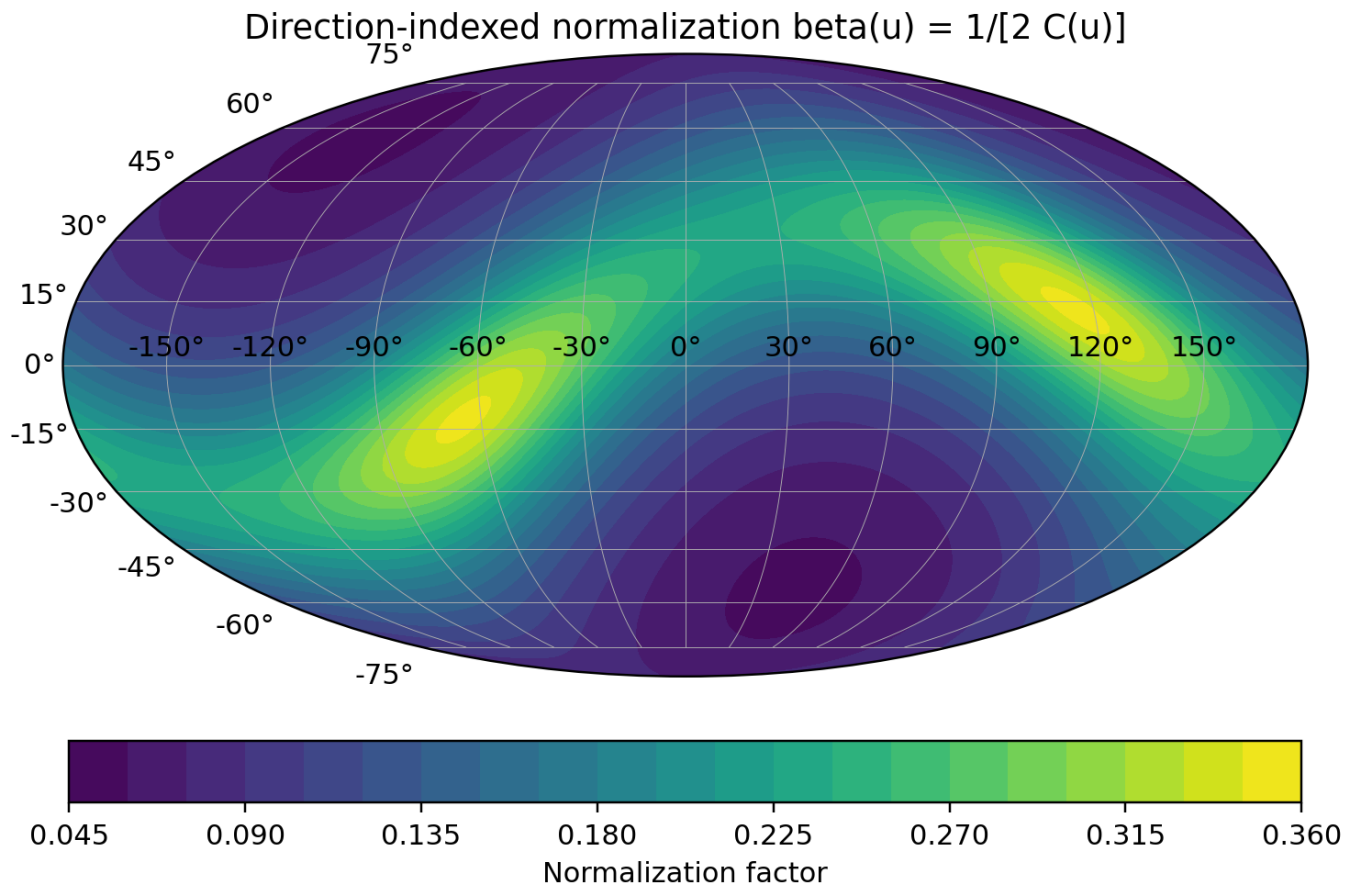


Figure 11. The direction-indexed normalization $\beta(u)$. It is positive over the complete sphere and antipodally even. Multiplying $\beta(u)$ by the corresponding raw phase cost returns one-half for every direction.

Normalization bounds

$$0.056816009673 \leq \beta(u) \leq 0.349814553706.$$

These bounds are finite because the raw metric has no zero eigenvalue. A null raw direction would require an infinite normalization and would break the self-indexed energy law. The positive spectral gap prevents that failure.

10. Interior a/b/c interaction structure

The raw phase scalar contains three diagonal channel terms and three symmetric interference terms:

$$\begin{aligned} \Phi &= W_{aa} \dot{a}^2 + W_{bb} \dot{b}^2 + W_{cc} \dot{c}^2 \\ &+ 2 W_{ab} \dot{a} \dot{b} \\ &+ 2 W_{ac} \dot{a} \dot{c} \\ &+ 2 W_{bc} \dot{b} \dot{c}. \end{aligned}$$

For Earth-CMB traversal, the ordered contributions are:

Interaction	Contribution (J/kg)	Contribution (GJ/kg)	Share of total
aa	4.099605354011e+11	409.960535401	+139.888700671%
bb	4.324266889894e+10	43.242668899	+14.755470938%
cc	1.578134858801e+09	1.578134859	+0.538498748%
2ab	-1.524129387169e+11	-152.412938717	-52.007074148%
2ac	-2.357169930606e+10	-23.571699306	-8.043248322%
2bc	1.426523555806e+10	14.265235558	+4.867652114%
Total	2.930619366940e+11	293.061936694	100.000000000%

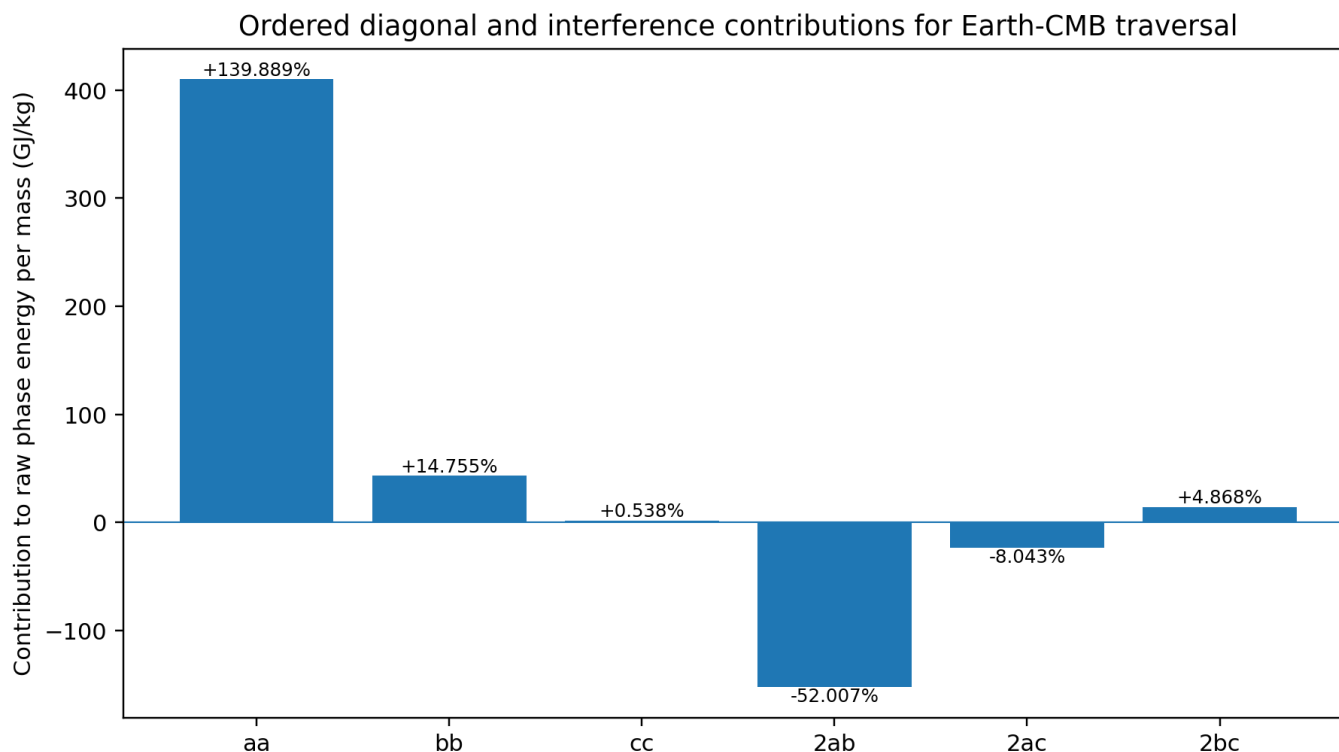


Figure 9. Diagonal and cross-channel contributions to the Earth-CMB raw phase scalar. Negative interference terms reduce the much larger positive aa contribution, while the complete quadratic form remains strictly positive.

This decomposition shows why plane-crossing count alone is insufficient. The energy is a coherent three-channel state. Individual ordered terms are basis components and may be positive or negative; only the complete contraction $\dot{q}^T W \dot{q}$ is invariant under a coordinated basis transformation.

11. Derivation of linear inertia

The complete derivation can now be stated in one sequence. A physical velocity selects a projective direction and speed. The direction fixes the ratios among $\dot{q}_a$, $\dot{q}_b$, and $\dot{q}_c$. The speed scales all three rates linearly. The coupling matrix produces a positive quadratic phase scalar. The self-indexed projective state supplies the exact W-to-M normalization. The resulting normalized scalar is $v^2/2$. Multiplication by mass gives kinetic energy.

```

v  -> qdot = 2 B^-1 v
qdot -> Phi = qdot^T W qdot
[qdot] -> beta = 1/[8 chi([qdot])]
E = m beta Phi = (1/2) m v^2.

```

Rule 1 - No null traversal direction

Every nonzero physical direction activates a nonzero Cell3 rate state. Positive definiteness of W and G_{phase} then guarantees positive total phase cost. The unit sphere is completely covered by positive-cost projective states.

Rule 2 - Kinetic energy is a rate-state energy, not traversal friction

Uniform $\dot{q}$ is a stationary coherent phase-flow state. Energy is stored in the rate state; it is not consumed in proportion to path length. Work is required to establish or change $\dot{q}$. Once established in a translation-invariant registry, the state persists without a drag term proportional to distance traveled.

Rule 3 - The speed-squared law is structural

The map $\dot{q} = 2 B^{-1} v$ is linear in velocity. The phase scalar is quadratic in $\dot{q}$. Therefore phase energy is quadratic in speed. The familiar v^2 dependence is not imposed after the fact; it follows from a linear traversal map combined with a bilinear interior interaction.

Rule 4 - Inertia is the curvature of the normalized phase-energy basin

For a fixed mass, the normalized energy is $E(v) = m v^2/2$. Its Hessian with respect to velocity is m times the identity. In the raw phase representation, the Hessian is directionally weighted by G_{phase} . Self-indexed normalization converts that raw curvature into the isotropic physical inertial tensor.

$$\partial^2 E / \partial v_i \partial v_j = m \delta_{ij}.$$

Rule 5 - Mass is the phase participation scale

The geometry and normalized directional law determine energy per unit mass. The material mass multiplies that energy. Within the traversal interpretation, mass is the extensive scale that measures how strongly the material state participates in the normalized Cell3 phase-energy basin. Doubling mass doubles energy, momentum, and force at fixed kinematics without changing the direction index.

Rule 6 - Antipodal symmetry separates energy from directed response

Velocity reversal maps $\dot{q}$ to $-\dot{q}$. The quadratic energy remains unchanged, while its first derivative reverses. Thus energy is even under antipodal reversal and momentum is odd. The same parity separation explains why a spin reversal can reverse a lateral force while leaving the rotational self-energy unchanged.

Rule 7 - Translation invariance produces persistent linear inertia

The same local Cell3 displacement dictionary repeats at every lattice center. The normalized traversal law depends on $\dot{q}$ and its projective direction, not on an absolute local cell number. A freely translating mass therefore carries its inertial phase-rate state through successive closure basins.

12. Momentum, force, and work from the traversal energy

Momentum is the gradient of kinetic energy with respect to physical velocity:

$$p = \partial E / \partial v = m v.$$

In address-rate coordinates, using $E = m \dot{q}^T M \dot{q} / 8$,

$$p_{i_q} = \partial E / \partial \dot{q} = (m/4) M \dot{q}.$$

The physical and address momenta are related by the transpose of the velocity Jacobian. Differentiating with time gives the force law.

$$\begin{aligned} F &= dp/dt = m a \\ d p_{i_q}/dt &= (m/4) M \ddot{q}. \end{aligned}$$

Work is the change in the normalized phase-energy state:

$$dE = p^T dv = m v^T dv.$$

A force therefore changes the Cell3 rate state. Constant velocity corresponds to constant $\dot{q}$ and constant energy. Acceleration is not motion through more cells; it is a change in the rate and phase relation with which cells are traversed.

Common-mode and relative motion

For a material point with local velocity v_R superposed on Earth-CMB translation V_E , the normalized Euclidean kinetic energy expands as one-half the squared total velocity. In a closed local system, the large common-mode baseline is shared. Observable local work ordinarily depends on changes and relative motions. The K-Field lateral reduction identifies a small channel by which part of the raw anisotropic mixed interaction does not close internally.

13. The lateral interaction as the mixed phase term

Let the total address rate be the sum of Earth translation and imposed local motion:

$$\dot{q}_{\text{total}} = \dot{q}_E + \dot{q}_R.$$

The raw phase scalar expands exactly as

$$\begin{aligned} \Phi_{\text{total}} &= \dot{q}_E^T W \dot{q}_E \\ &+ \dot{q}_R^T W \dot{q}_R \\ &+ 2 \dot{q}_R^T W \dot{q}_E. \end{aligned}$$

Term	Physical interpretation
$\dot{q}_E^T W \dot{q}_E$	Common Earth-CMB linear traversal baseline
$\dot{q}_R^T W \dot{q}_R$	Local rotational or oscillatory self-energy
$2 \dot{q}_R^T W \dot{q}_E$	Translation-local-motion interference term

The single-cell reduction identifies the mixed scalar in both coordinate systems:

$$v_R^T G_{\text{phase}} V_E = \dot{q}_R^T W \dot{q}_E.$$

Its coordinate-differential energy and acceleration kernel is

$$\begin{aligned} \epsilon_{\text{cross}} &= \kappa_F \dot{q}_R^T W \dot{q}_E \\ a &= -(\kappa_F/\rho) u [\dot{q}_R^T W \dot{q}_E]. \end{aligned}$$

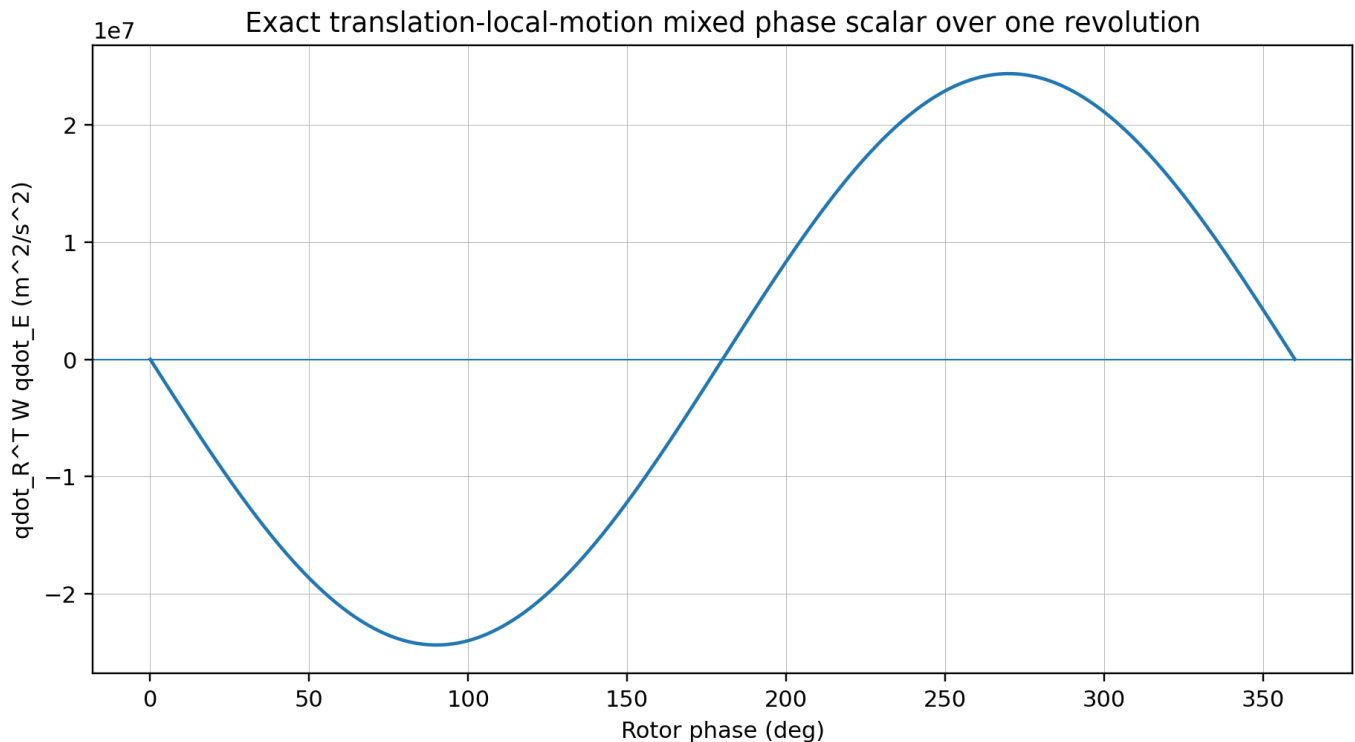


Figure 12. The exact mixed scalar over one revolution at the supplied outer tube radius. The waveform is generated from the supplied $\dot{q}_R$ sine and cosine coefficient vectors and the supplied $W \dot{q}_E$ vector. Positive and negative portions are differential phase modulation; total baseline crossing rates remain one-signed.

Linear inertia and lateral response are therefore different sectors of the same quadratic traversal form. The baseline self-term carries linear kinetic energy. The local self-term carries rotational or oscillatory energy. The mixed term is ordinarily internal to the total energy, but the lateral kernel extracts a small directed residual from it.

14. Quantitative lateral coupling and channel cancellation

For the Earth-CMB direction, the full normalized coefficient multiplying the raw mixed scalar in the expansion of kinetic energy is $2 \beta_E = 0.4666634234$. The supplied lateral coefficient is $\kappa_F = 3.0740508997e-11$. Their ratio is

$$S_{\text{perp}} = \kappa_F / (2 \beta_E) = 6.587297709194528e-11.$$

Thus the expressed lateral channel is approximately $6.5873e-11$ of the full direction-normalized mixed interaction under the supplied equations. This is a dimensionless comparison of coefficients. It does not by itself identify the dimensional gravitational constant; a separate density-time or equivalent dimensional bridge is required for that identification.

Rotation channel	Mean acceleration vector (m/s ²)	Signed lateral component (m/s ²)	Share of lateral output
a	0.002624082468, 0.048335395280, 0.043760877443	0.065204463511	87.042128933%
b	-0.010491506362, 0.007993321910, 0.012203904109	0.013214227551	17.639812312%
c	0.013670777762, -0.000341756842, -0.006533042570	-0.003507306988	-4.681941245%

Mean acceleration branch vectors and exact resultant

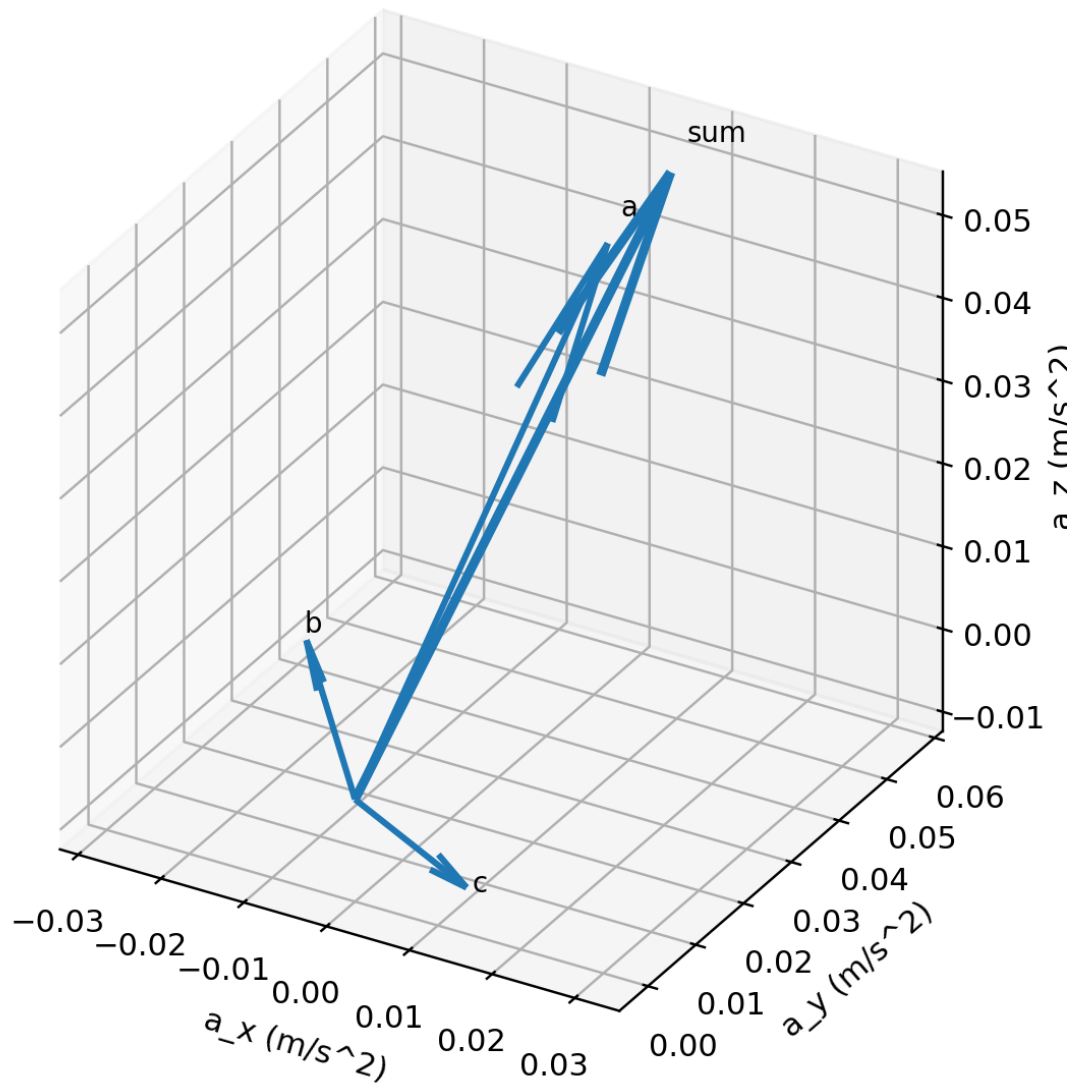


Figure 13. Mean acceleration vectors from the a, b, and c rotation branches. Their nonlateral components cancel in the ordered sum, leaving the supplied resultant. Individual branch vectors are address-basis components, not independent invariant forces.

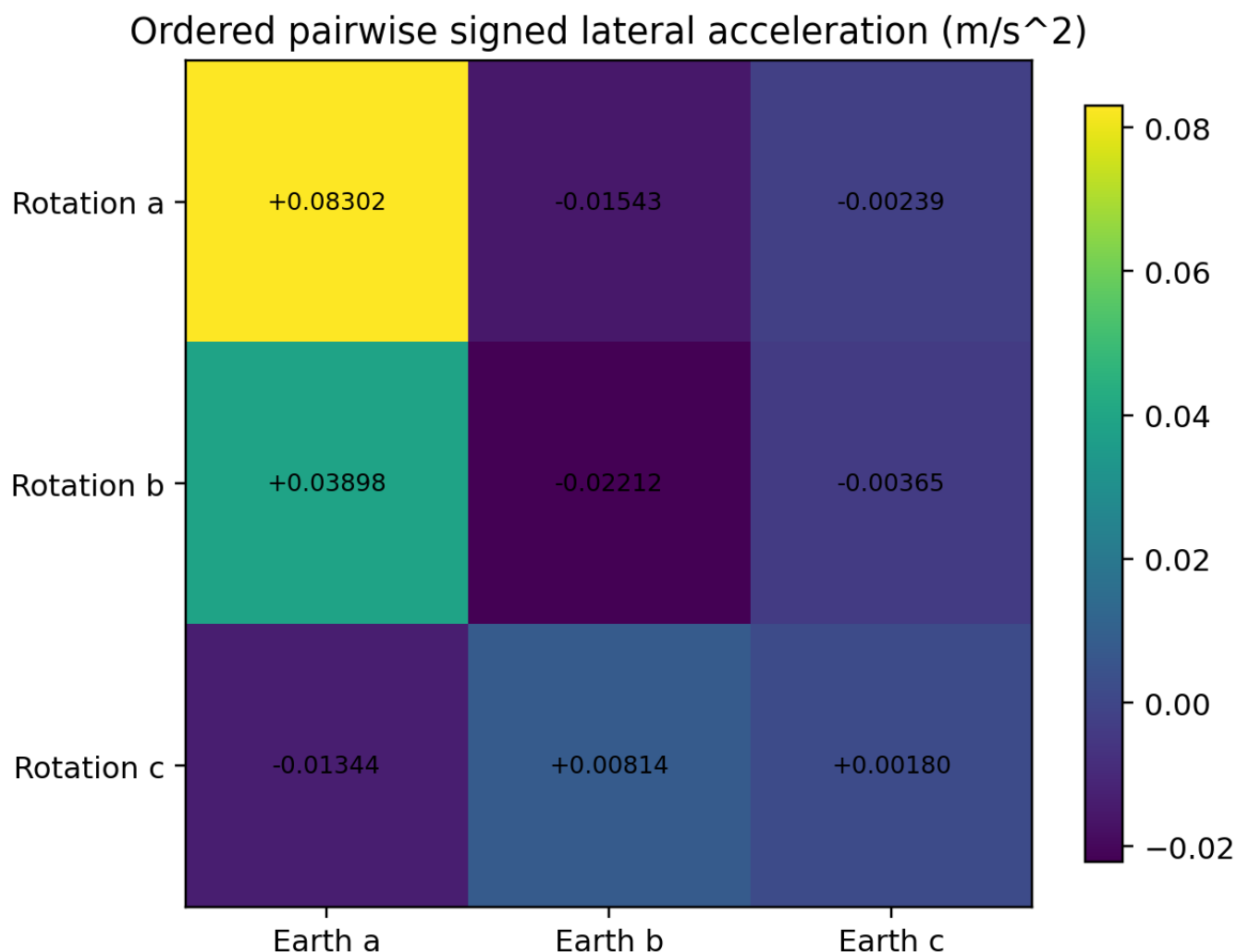


Figure 14. Ordered 3 x 3 decomposition by rotation channel and Earth traversal channel. The entries are signed lateral components in m/s^2 . The matrix displays strong cancellation and reinforcement across the coupled basis.

The dominant a rotation branch contributes about 87.0421 percent of the final lateral output, the b branch contributes 17.6398 percent, and the c branch contributes -4.68194 percent. Their signed shares sum to 100 percent. This is another manifestation of coherent channel interaction: a negative branch does not invalidate positive total energy; it participates in the vector cancellation that isolates the lateral residual.

15. The rotating-tube single-cell witness

The source experiment uses a stainless-steel tube with outer radius 5 mm, inner radius 4 mm, length 0.1 m, mass 0.0226194671 kg, spin rate 50,000 rpm, angular speed 5235.987756 rad/s, and rotational energy 6.35628672 J. During one 0.0012 s revolution, Earth-CMB translation advances the material by 443.7750024 m. The absolute path is therefore a translated circular or generalized screw-trochoidal path rather than a simple helix about the spin axis.

Quantity	Value
Tube mass	0.022619467105847 kg
Axial moment of inertia	4.636990756698536e-07 kg m ²
Spin angular speed	5235.987755982988 rad/s
Spin energy	6.356286719461 J
Translation per revolution	443.775002400488 m
Mean acceleration magnitude	0.074911384073992 m/s ²

Quantity	Value
Force magnitude	1.694455587915101e-03 N
Cycle velocity increment	8.989366088879059e-05 m/s

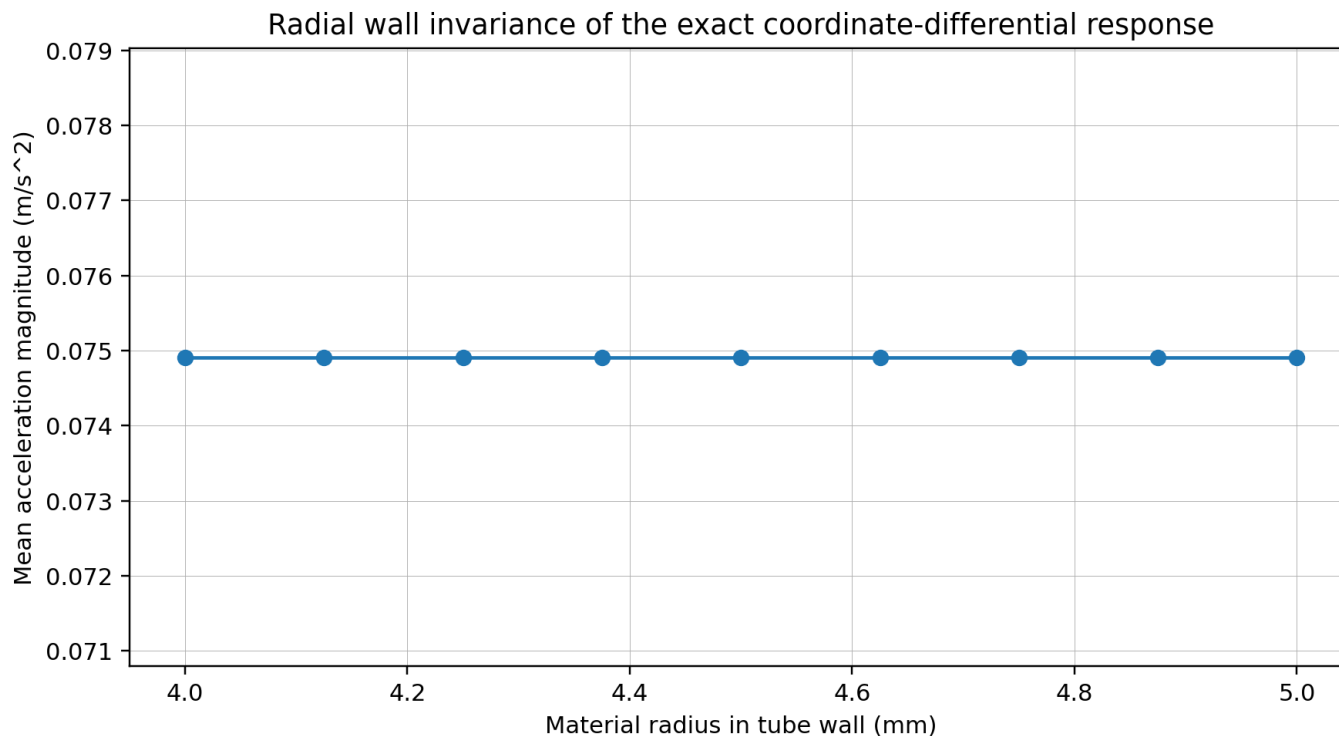


Figure 15. The exact mean acceleration magnitude evaluated through the actual 4-5 mm tube wall. Address-rate amplitude changes linearly with radius while velocity increment per unit address change scales inversely, leaving the integrated mean acceleration invariant to numerical precision.

The radial invariance is important for the inertia interpretation. It shows that the coordinate-differential kernel is not an artifact of replacing the distributed wall by one inertia-equivalent radius. The source uses the full physical wall, while the inertia-equivalent radius is retained only as an exact surrogate for total rotational energy and angular momentum.

16. Consequences for a self-indexing physical space

The derived relationship joins address identity and energy identity. Every nonzero direction has a positive raw phase cost. The self-indexed projective state identifies which directional gain applies. The direct metric supplies physical speed. The coupling metric supplies interior phase interaction. Their ratio returns ordinary kinetic energy. The same architecture therefore answers both where a state is directed and how much normalized traversal energy it carries.

Positive cost does not imply drag

A positive phase cost is stored energy, not a dissipative loss. A mass moving uniformly does not pay the same energy again at every plane. Its coherent $\dot{q}$ state is propagated through the translation-invariant registry. Dissipation would require an additional irreversible term, such as dependence on path length, entropy production, or a velocity-proportional force. None is present in the quadratic conservative form derived here.

Continuum motion through a discrete registry

The address lattice is discrete, while a measured material trajectory is continuous. The source establishes an exact coordinate-differential kernel. Continuous motion can therefore be represented as changing phase within closure basins and coherent transitions between indexed states. The present derivation does not require every infinitesimal energy increment to be localized at a node or boundary; it requires only the exact rate-state map and the quadratic closure identity.

Why the unit sphere is fully indexed

Primitive integer rays form a countable dense directional registry as address magnitude increases. Between exact primitive rays, the continuous witness remains associated with a closure basin and a projective rate ratio. Since G_{phase} is positive definite on the continuous sphere, every such direction has positive cost. Self-indexing and positive definiteness therefore complement each other: indexing supplies identity; the phase metric supplies nonzero energy curvature.

17. Experimental and computational pathways

The framework produces specific measurement programs that distinguish baseline self-energy, local self-energy, and mixed response.

Program	Predicted relationship
Directional raw-gain mapping	Measure a phase-sensitive observable while sweeping a controlled velocity or oscillation direction. The internal raw response should follow $C(u)$, bounded by the three supplied eigenvalues, while normalized ordinary inertia remains isotropic.
Spin-reversal parity	Reverse local rotation at fixed apparatus orientation. The local self-energy remains even, while the mixed lateral term reverses sign.
Axis sweep	Sweep the spin axis over the unit sphere. The lateral envelope follows the cross-product geometry involving G_{phase} V_E , with extrema fixed by the supplied tensor and Earth-CMB vector.
Mass scaling	At fixed geometry and velocity, kinetic energy, momentum, and force scale linearly with participating mass, while $C(u)$, $\beta(u)$, and channel ratios remain unchanged.
Acceleration transient	Apply a controlled change in velocity. The energy change follows $m \mathbf{v} \cdot d\mathbf{v}$ and the address momentum follows $(m/4) \mathbf{M} \cdot d\mathbf{v}$.
Plane-localization test	Use structures whose dimensions approach the Cell3 spacings. Search for phase-locked or boundary-localized signatures that distinguish continuous bulk interaction from discrete plane or node localization.
High-frequency coherence test	Increase local traversal frequency at fixed rim speed and geometry. A departure from the low-frequency quadratic kernel would locate a coherence or dispersion transition.

The first five programs test closed algebraic consequences of the current model. The final two address the microscopic composition rule that remains outside the continuous reduction.

18. Localization boundary and model closure

The source reduction explicitly establishes an exact coordinate-differential kernel but does not localize the impulse exclusively at nodes, face crossings, plane transitions, or one particular interior substructure. That boundary must remain exact in an archival derivation. The present paper therefore distinguishes three levels:

Level	Established content
Geometric level	Physical B, direct and reciprocal metrics, 27 nodes, three plane families, spacings, normals, and self-indexed projective directions
Rate-energy level	$\dot{q} = 2 B^{-1} \mathbf{v}$; positive W and G_{phase} ; raw phase scalar; direction-indexed normalization; kinetic energy, momentum, force, and mixed interaction
Microscopic localization level	A composition law specifying where within a cell the continuous kernel is physically concentrated or exchanged

The first two levels are sufficient to derive linear inertia as a coherent traversal-rate energy. The third level is required to identify the microscopic carrier and the precise event topology of energy exchange. No exclusive localization rule is inserted into this report without source support.

19. Findings and conclusion

No.	Finding
1	Physical velocity maps exactly and invertibly into three Cell3 half-step traversal rates.
2	The address coupling W and physical response G_{phase} are strictly positive definite.
3	Every nonzero unit-sphere direction therefore has positive raw phase cost; no null direction exists.
4	The raw directional phase cost is antipodally even and bounded by the eigenvalues of G_{phase} .
5	Self-indexing supplies the occupied projective direction and the exact direction-dependent normalization $\beta(u)$.
6	The normalized phase scalar equals ordinary kinetic energy per unit mass, $v^2/2$, in every direction.
7	Along Earth-CMB traversal, 293.0619367 GJ/kg of raw phase scalar normalizes to 68.3806433 GJ/kg of kinetic energy.
8	Linear inertia is the resistance to changing the normalized coherent a/b/c traversal-rate state.
9	Momentum is the first derivative and inertial mass is the curvature scale of this normalized energy state.
10	The rotating-mass lateral response is the residual mixed term between Earth baseline traversal and local traversal.
11	The supplied lateral coefficient expresses approximately $6.5873e-11$ of the full normalized mixed interaction.
12	The continuous rate-energy derivation is closed; exclusive node or plane-crossing localization remains a separate composition rule.

The resulting relationship is compact:

```
E_kinetic = self-index-normalized positive Cell3 traversal phase energy.

Linear inertia = resistance to changing the coherent traversal-rate state.

Lateral response = residual of the translation-local-motion mixed phase term.
```

This architecture unifies linear kinetic energy, inertial momentum, force response, rotational coupling, and directional phase modulation without introducing a zero-cost path or a direction outside the self-indexed sphere.

Appendix A. Authoritative matrices

Direct metric M (nm ²)	col 1	col 2	col 3
row 1	0.111676057905	0.0207103332816	0.0439372350978
row 2	0.0207103332816	0.343877732397	0.296392758962
row 3	0.0439372350978	0.296392758962	0.627770610069

Reciprocal metric R (nm ⁻²)	col 1	col 2	col 3
row 1	9.20802552868	0.00153480242812	-0.645188035677
row 2	0.00153480242812	4.90339831231	-2.31517558442
row 3	-0.645188035677	-2.31517558442	2.73117127475

Address coupling W (nm ²)	col 1	col 2	col 3
row 1	0.0833533026335	0.137500576604	0.102559211222
row 2	0.137500576604	0.692400042849	0.550800904453
row 3	0.102559211222	0.550800904453	0.587747652353

Physical response G_phase	col 1	col 2	col 3
row 1	2.9764106378	1.2431507795	-2.0803600354
row 2	1.2431507795	2.5922286635	-2.3178839937
row 3	-2.0803600354	-2.3178839937	6.7735709133

Matrix identity

$$\max |G_{\text{phase}} - 4 B^{(-T)} W B^{(-1)}| = 2.220446e-15.$$

Appendix B. Complete 27-node coordinate ledger

Address L	Kind	lambda	Polarity	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)
(-1, -1, -1)	corner	-3	-1	-0.176351350143518	0.118682429904309	0.637311498325592
(-1, -1, 0)	edge center	-2	1	-0.181416008956851	-0.076211421277593	0.292444294144131
(-1, 0, -1)	edge center	-2	1	-0.158685962221493	0.222342355125059	0.363610955629613
(0, -1, -1)	edge center	-2	1	-0.012600729108693	0.091233925961153	0.618567746877440
(-1, -1, 1)	corner	-1	-1	-0.186480667770183	-0.271105272459495	-0.052422910037330
(-1, 0, 0)	face center	-1	-1	-0.163750621034825	0.027448503943156	0.018743751448153
(-1, 1, -1)	corner	-1	-1	-0.141020574299467	0.326002280345807	0.089910412933635
(0, -1, 0)	face center	-1	-1	-0.017665387922025	-0.103659925220749	0.273700542695978
(0, 0, -1)	face center	-1	-1	0.005064658813332	0.194893851181902	0.344867204181461
(1, -1, -1)	corner	-1	-1	0.151149891926133	0.063785422017997	0.599823995429287
(-1, 0, 1)	edge center	0	1	-0.168815279848158	-0.167445347238745	-0.326123452733309
(-1, 1, 0)	edge center	0	1	-0.146085233112800	0.131108429163905	-0.254956791247826
(0, -1, 1)	edge center	0	1	-0.022730046735358	-0.298553776402651	-0.071166661485483
(0, 0, 0)	body center	0	1	0.000000000000000	0.000000000000000	0.000000000000000
(0, 1, -1)	edge center	0	1	0.022730046735358	0.298553776402651	0.071166661485483
(1, -1, 0)	edge center	0	1	0.146085233112800	-0.131108429163905	0.254956791247826
(1, 0, -1)	edge center	0	1	0.168815279848158	0.167445347238745	0.326123452733309
(-1, 1, 1)	corner	1	-1	-0.151149891926133	-0.063785422017997	-0.599823995429287
(0, 0, 1)	face center	1	-1	-0.005064658813332	-0.194893851181902	-0.344867204181461
(0, 1, 0)	face center	1	-1	0.017665387922025	0.103659925220749	-0.273700542695978
(1, -1, 1)	corner	1	-1	0.141020574299467	-0.326002280345807	-0.089910412933635
(1, 0, 0)	face center	1	-1	0.163750621034825	-0.027448503943156	-0.018743751448153
(1, 1, -1)	corner	1	-1	0.186480667770183	0.271105272459495	0.052422910037330
(0, 1, 1)	edge center	2	1	0.012600729108693	-0.091233925961153	-0.618567746877440
(1, 0, 1)	edge center	2	1	0.158685962221493	-0.222342355125059	-0.363610955629613
(1, 1, 0)	edge center	2	1	0.181416008956851	0.076211421277593	-0.292444294144131
(1, 1, 1)	corner	3	-1	0.176351350143518	-0.118682429904309	-0.637311498325592

Appendix C. Earth-CMB and normalized energy ledger

Quantity	Value
Earth-CMB vector V_E (m/s)	-359003.199065813038033, 76679.886034001989174, -44701.060100687995146
Speed (m/s)	369812.502000406442676
Address rate qdot_E (s^-1)	-2.217734842862322e+15, 2.499065863112239e+14, 5.181751775713255e+13
G_phase V_E (m/s)	-880221.981635091360658, -143911.436561599461129, 266335.026948619284667
Raw phase scalar (J/kg)	2.930619366939656e+11
Ordinary kinetic energy (J/kg)	6.838064331790031e+10
Directional cost C_E	2.142872035668970
Address metric ratio chi_E	0.535718008917242
Normalization beta_E	0.233331711682871
Lateral coefficient kappa_F	3.074050899702035e-11
Lateral/full mixed ratio	6.587297709194528e-11

Appendix D. Source lineage and reproducibility

Source file	SHA-256	Role
K-Field_single_cell3_lateral_force_reduction_1784985105.json	62e4472aacc7df503cc4b5120d7abce97589891e8d22843177ee547e35b71a6f	Physical nanometre Cell3 geometry, rate mapping, Earth-CMB vector, W, G_phase, tube trajectory, and lateral reduction
K-Field_self-indexing_spectral_fabric_1784510144(1).json	5c8d9147b4f0e149716122f23620380ecbaef92d6fb6510e97a3c7c657609726	Self-indexing projective channels, translation invariance, radial addressing, and even/odd spectral phase structure
K-Field_functionally_immune_to_non-coherent_noise_in_address_determination_1784513477(1).json	5d877d4899b06cf95a57528380376b01f7159077662c817670778f549ef63569	Closure basins, redundant direct/reciprocal constraints, and stable address determination

All numerical figures in this report were generated directly from the listed source values. No external illustrative geometry or synthetic measurement data were introduced. The unit-sphere maps are exhaustive numerical evaluations of the supplied tensor on a regular angular grid. The finite radial codebook is generated from the supplied native Gram metric over the explicitly stated bounded integer domain.

Appendix E. Numerical verification contract

Check	Residual or result
G_identity_max_abs_error	1.7763568394002505e-15
direct_metric_max_abs_error	0.0
reciprocal_metric_max_abs_error	0.0
volume_error_nm3	0.0
node_source_max_error_nm	0.0
face_area_max_error_nm2	0.0
face_spacing_max_error_nm	5.551115123125783e-17
mean_acceleration_source_error_m_s2	0.0
cycle_endpoint_error_m_s	1.9548668967155818e-19
pairwise_sum_error_m_s2	2.0461098328902598e-16
wall_integration_error_m_s2	5.605063087764203e-17
total_rate_direction_reversal_count	0
Derived M identity max residual (m^2)	9.629650e-35
Derived G identity max residual	2.220446e-15
Physical/address raw phase equality residual (J/kg)	1.220703e-04
Normalized/classical energy residual (J/kg)	0.000000e+00
Minimum eigenvalue W (nm^2)	0.050269586792171
Minimum eigenvalue G_phase	1.429328753487483

Appendix F. Symbol glossary

Symbol	Meaning
a, b, c	Physical Cell3 basis vectors and corresponding address channels.
B	3×3 matrix with columns a, b , and c .
q	Dimensionless centered half-step address coordinate.
$\dot{q}$	Cell3 address traversal rate in s^{-1} .
M	Direct metric $B^T B$.
R	Reciprocal metric M^{-1} .
W	Address-space phase coupling matrix.
G_{phase}	Dimensionless physical phase-response tensor $4 B^T W B^{-1}$.
u	Physical unit direction.
$C(u)$	Raw phase cost $u^T G_{\text{phase}} u$.
χ	Projective W -to- M metric ratio.
$\beta(u)$	Direction-indexed normalization $1/[2 C(u)]$.
Φ	Raw phase scalar $\dot{q}^T W \dot{q} = v^T G_{\text{phase}} v$.
V_E	Earth-CMB traversal velocity.
$\dot{q}_E$	Earth-CMB address-rate vector.
$\dot{q}_R$	Locally imposed rotational or oscillatory address-rate vector.
κ_F	Supplied lateral mixed-term coupling coefficient.
closure basin	Continuous witness region decoding to one exact self-indexed address.